

Yousef Nejatbakhsh, Ph.D.

MODEL RISK MANAGEMENT | QUANTITATIVE ANALYTICS | DATA SCIENCE | APPLIED AI

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Google Scholar: scholar.google.com/citations?user=kd0YBMkAAAJ

Website: <http://127.0.0.1:5500/index.html#home>

GitHub: <https://github.com/dashboard>

Public GitHub: github.com/Nejatbakhsh-y

PROFESSIONAL PROFILE

Ph.D.-trained applied mathematician, data scientist, and model-risk/quantitative analytics professional with 16+ years of experience across statistical modeling, machine learning, optimization, forecasting, risk analytics, model evaluation, documentation, and analytical communication. Recent portfolio work includes end-to-end independent validation frameworks for central-counterparty margin and FICC-style Treasury-clearing liquidity stress, together with governance-oriented validation of high-stakes financial machine-learning models. Skilled in conceptual soundness review, champion/challenger benchmarking, backtesting, sensitivity and stress testing, implementation verification, data-quality controls, fairness, calibration, explainability, drift monitoring, and governance-ready reporting. Proven ability to translate technical evidence into clear conclusions, limitations, findings, remediation priorities, and executive-level decision support.

CORE COMPETENCIES

Model Risk Management and Validation	Independent review, effective challenge, conceptual soundness, model lifecycle documentation, performance monitoring, assumption and limitation review, issue tracking, remediation narratives, model inventory, and SR 11-7 awareness.
Financial Risk and Quantitative Analytics	CCP margin, clearing liquidity, Treasury and repo risk, stress/scenario analysis, LCR and shortfall analysis, credit and bankruptcy risk, market-risk interpretation, forecasting, and portfolio analytics.
Statistical Modeling and Machine Learning	Regression, logistic regression, GLMs, classification, clustering, random forests, gradient boosting, neural networks, cross-validation, calibration, ROC/AUC, precision/recall, F1, MSE/RMSE/MAE, and threshold logic.
Responsible AI and AI Governance	Champion/challenger review, fairness and bias testing, explainability, explanation stability, drift monitoring, model cards, governance scores, committee reporting, GenAI/RAG evaluation, and hallucination-risk analysis.
Data Controls and Reproducibility	Python/SQL/R workflows, data lineage, source-to-output traceability, validation flags, deterministic seeds, configuration control, independent recalculation, unit and integration testing, CI quality gates, and evidence generation.
Communication, Leadership, and Coaching	Validation reports, executive summaries, technical findings, model limitations, stakeholder presentations, cross-functional collaboration, former department chair, and mentorship of 30+ graduate theses.

TECHNICAL SKILLS

Programming and Data	Python, SQL, R, MATLAB, pandas, NumPy, SciPy, scikit-learn, statsmodels, PyTorch, Transformers, Jupyter, Google Colab, Excel, LaTeX, Git, and GitHub.
Model Validation and Risk	Conceptual soundness, independent validation, benchmarking, challenger models, backtesting, sensitivity analysis, historical/hypothetical/reverse stress testing, implementation verification, procyclicality, findings management, and ongoing monitoring.
Financial and Quantitative Methods	Historical-simulation VaR, parametric/EWMA models, margin add-ons, liquidity cash-flow modeling, Cover 1/Cover 2, LCR, Treasury duration/convexity, repo rollover, collateral haircuts, settlement fails, credit risk, risk scoring, time-series forecasting, and portfolio optimization.
Machine Learning and Responsible AI	Linear/logistic models, tree ensembles, SVM concepts, clustering, neural networks, LSTM, Transformer attention, recommender systems, calibration, fairness metrics, SHAP/permutation importance, drift metrics, responsible AI, prompt engineering, and RAG evaluation.
Data Engineering and Delivery	DuckDB, Parquet, YAML configuration, relational data concepts, SQL joins/CTEs/aggregations, ETL/ELT, data dictionaries, reproducible pipelines, GitHub Actions, pytest, coverage, Ruff, mypy, Streamlit, Power BI, Tableau, and dashboard QA.
Reporting and Governance	Independent validation reports, model cards, AI governance cards, committee-ready summaries, technical documentation, source contracts, data-quality evidence, limitation registers, risk findings, KPI reporting, and stakeholder presentations.

SELECTED GITHUB PROJECTS

- FICC Treasury Clearing Liquidity Stress Testing and Model Validation | *Independent model-validation research framework*

- Developed a reproducible Python framework using public Federal Reserve and New York Fed data, including FR 2004 primary-dealer statistics, SOFR, H.15 Treasury yields, and H.4.1 system-liquidity indicators; classified all material fields as observed, derived, synthetic, assumed, or modeled.
 - Implemented synthetic clearing-member portfolios, configurable default sets, baseline liquidity cash flows, Treasury yield shocks, repo funding stress, collateral-haircut stress, settlement-fail stress, integrated stressed-liquidity requirements, Cover 1/Cover 2 analysis, reverse stress testing, LCR, shortfalls, and resource-utilization diagnostics.
 - Built independent-validation evidence for conceptual soundness, separate implementation verification, and multi-parameter sensitivity analysis, with deterministic configuration, SQL/data-quality controls, pytest, Ruff, mypy, and GitHub Actions quality gates.

2026
- CCP Margin Model Independent Validation Framework | *Completed model-risk-management portfolio project*

- Built an end-to-end central-counterparty margin validation framework with deterministic synthetic member portfolios, public market and macroeconomic data, a 99% historical-simulation VaR primary model, a parametric EWMA challenger, liquidity/concentration/gap-risk add-ons, and a configurable stress buffer.
 - Executed model challenge through Kupiec and Christoffersen backtesting, Basel-style traffic-light assessment, margin-shortfall testing, benchmark comparison, sensitivity analysis, historical/hypothetical/reverse stress testing, procyclicality assessment, and independent recalculation.
 - Produced reproducible evidence, validation findings, DuckDB analytical tables, automated tests, CI controls, and an executive Streamlit dashboard for model performance, stress, sensitivity, and governance reporting.

2026
- AssuranceTwin AI - Model Validation and Governance | *High-stakes financial machine-learning governance*

- Created a reproducible independent-validation workflow for public mortgage-lending data, comparing logistic regression, random forest, gradient boosting, calibrated models, and champion/challenger alternatives across predictive and governance criteria.
 - Evaluated fairness and bias, calibration, explainability and explanation stability, adverse stress scenarios, out-of-time performance, population and characteristic stability, prediction/performance/fairness drift, and monitoring readiness.
 - Generated a model inventory, model card, AI governance card, composite AssuranceTwin governance score, model-risk committee report, validation findings, and a Streamlit governance dashboard.

2026
- FedNarrativeAlpha: GenAI-Assisted Alpha Research Pipeline | *Liquid ETF quantitative research and model-risk controls*

- Developed a reproducible Python research platform integrating ETF prices, FRED macroeconomic indicators, Federal Reserve communication embeddings, information-theoretic feature selection, baseline and machine-learning models, and risk-constrained portfolio construction.
 - Implemented chronological walk-forward validation, no-lookahead and feature-date tests, transaction-cost-aware backtesting, benchmark comparison, portfolio-risk controls, model-card documentation, and reproducible research reporting.

2026

SELECTED APPLIED ANALYTICS AND RESEARCH EVIDENCE

Financial Risk and Failed-Bank Analytics	Developed a two-model Silicon Valley Bank bankruptcy and acquisition-readiness framework combining logistic regression with transparent weighted scoring; built interpretable indicators, threshold logic, model-comparison narratives, limitations, and decision-oriented outputs. Designed FDIC failed-bank and BankFind-style workflows with acquirer identifiers, transaction fields, validation flags, exclusion logic, and dashboard-ready financial features.
Hybrid Deep Learning for Financial Forecasting	Developed accepted IEEE Access research integrating LSTM, Transformer attention, federated learning, and sentiment analysis for stock-market prediction, emphasizing architecture selection, feature integration, evaluation metrics, methodological limitations, and reproducibility.
GenAI, RAG, and AI Evaluation	Designed multi-document RAG evaluation concepts covering benchmark questions, prompt variation, ambiguity, temporal conflicts, retrieval quality, groundedness, completeness, citation support, abstention analysis, and hallucination-risk assessment.
Healthcare Recommender Systems and Optimization	Developed collaborative/content-based recommendation concepts for phenylketonuria nutrition support and contributed to sustainable supply-chain optimization research incorporating routing, carbon emissions, scenario analysis, and multi-period decision models.

PROFESSIONAL EXPERIENCE

- Full-Time Lecturer, Department of Mathematical Sciences | Kean University | Union, NJ

2024 - 2026

- Lead quantitative, statistical, data-visualization, and applied analytics instruction with emphasis on problem framing, data preparation, model interpretation, assumptions, limitations, validation, and decision-oriented communication.
 - Developed a Data Visualization course for Kean University Online, aligning dashboard design, visual storytelling, analytical interpretation, and workforce-relevant skills.
 - Mentor machine-learning, AI, and analytics projects involving data collection, preprocessing, exploratory analysis, model development, evaluation, reproducibility, and stakeholder-ready reporting using Python, R, SQL concepts, and Google Colab.
 - Translate complex quantitative methods into controlled analytical workflows, strengthening communication and documentation standards relevant to model risk, AI governance, business intelligence, and risk-control functions.

Part-Time Lecturer, Department of Statistics | Rutgers University | New Brunswick, NJ

2022 - Present

- Deliver statistics and research methods instruction focused on probability, inference, regression, uncertainty, reproducibility, model interpretation, diagnostics, and evidence-based reasoning.
- Guide learners through problem framing, data preparation, exploratory analysis, validation, written interpretation, and communication of quantitative findings.
- Explain statistical assumptions, limitations, uncertainty, and actionable conclusions to mixed technical and nontechnical audiences.

Assistant Professor, Mathematics Department | Caldwell University | Caldwell, NJ

2022 - 2024

- Delivered Applied Statistics, Data Mining, Business Data Analysis, Data Visualization with Python, Data Visualization with R, Probability, and Statistics for Research using applied datasets and project-based analytics.
- Designed assignments requiring data cleaning, exploratory analysis, statistical modeling, model evaluation, visualization, written reporting, and presentation of findings.
- Advised and mentored quantitative research involving data analysis, computational modeling, forecasting, machine learning, and practical interpretation of statistical evidence.

Adjunct Professor, Business Statistics, Management Department | New Jersey Institute of Technology | Newark, NJ

2020 - 2023

- Delivered business statistics and quantitative decision-support instruction emphasizing Excel, statistical reasoning, data interpretation, and management-facing analytical communication.
- Prepared assessments requiring data validation, interpretation of business metrics, concise explanation of quantitative findings, and evidence-based recommendations.

Adjunct / Part-Time Faculty in Mathematics, Statistics, and Data-Oriented Courses | Kean University, Union College, Essex County College, and related institutions | NJ

2019 - 2024

- Supported mathematics, statistics, probability, data analysis, linear algebra, discrete mathematics, calculus, MATLAB, and computationally oriented quantitative courses in in-person, hybrid, and online formats.
- Used Canvas, Blackboard, Moodle, Pearson, WebAssign, Zoom, and Microsoft 365 to manage instruction, assessment workflows, learner support, and structured documentation.

Researcher / Quantitative Modeling Contributor | Stevens Institute of Technology and Rutgers Business School Research Groups | NJ

2016 - 2018

- Contributed to financial engineering, supply-chain modeling, optimization, scenario analysis, and risk-assessment research using applied mathematics and quantitative modeling.
- Presented invited talks on fuzzy-logic applications in financial engineering and supply-chain risk assessment, translating mathematical methods into applied decision-support contexts.

Assistant Professor, Department Chair, and Educational Administrator | Islamic Azad University | Firuzkuh / Tehran, Iran

2007 - 2016

- Directed applied mathematics, optimization, numerical analysis, machine learning, neural networks, fuzzy systems, and quantitative decision-modeling work while serving in faculty and academic leadership roles.
- Supervised more than 30 M.Sc. theses involving bank credit analysis, bankruptcy risk, forecasting, business intelligence, multi-criteria decision models, data mining, optimization, and financial analytics.
- Managed curriculum planning, faculty coordination, student advising, documentation, quality review, and leadership reporting.

EDUCATION

Ph.D.	Fuzzy Applied Mathematics, Islamic Azad University, Science and Research Branch, Tehran, Iran, 2010. Dissertation: Numerical Solution of Fuzzy Differential Inclusion.
M.Sc.	Applied Mathematics, Islamic Azad University, Science and Research Branch, Tehran, Iran, 2006. Thesis: Numerical Solution of Fuzzy Differential Equations.
B.Sc.	Pure Mathematics, Payam Noor University, Khoy, Iran, 2003.

SELECTED RESEARCH AND PUBLICATIONS

- Y. Nejatbakhsh and M. Aliasgari, "Enhancing Stock Market Prediction with Hybrid Deep Learning: Integrating LSTM, Transformer Attention, Federated Learning, and Sentiment Analysis," IEEE Access, accepted 2025.
- Y. Nejatbakhsh et al., "Recommending Healthy Food for Phenylketonuria with Collaborative and Content-Based Filtering," IEEE HealthCom, 2024.
- K. Forghani, R. Kia, and Y. Nejatbakhsh, "A Multi-Period Sustainable Hydrogen Supply Chain Model Considering Pipeline Routing and Carbon Emissions: The Case Study of Oman," Renewable and Sustainable Energy Reviews, 2023.
- Y. Nejatbakhsh and M. Aliasgari, "Private Distributed Coded Computation," Applications of Computer Algebra, ACA 2022.
- T. Allahviranloo, M. Shafiee, and Y. Nejatbakhsh, "A Note on Fuzzy Differential Equations and the Extension Principle," Information Sciences, 2009.

LANGUAGES AND ADDITIONAL INFORMATION

- Languages: Persian - Native/Bilingual; English - Fluent; Turkish - Fluent; Arabic - Intermediate; Spanish - Basic.
- Location and Availability: Based in Basking Ridge, NJ; available for hybrid work within a reasonable commuting distance of New York City and Jersey City.
- Work Authorization: Authorized to work in the United States.